TITLE PAGE

- **Problem Statement ID –**SIH26102
- **Problem Statement Title-**Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.
- **Theme-BlockChain&Cybersecurity**
- **PS Category-** Software
- **Team Name –** Code Dominators



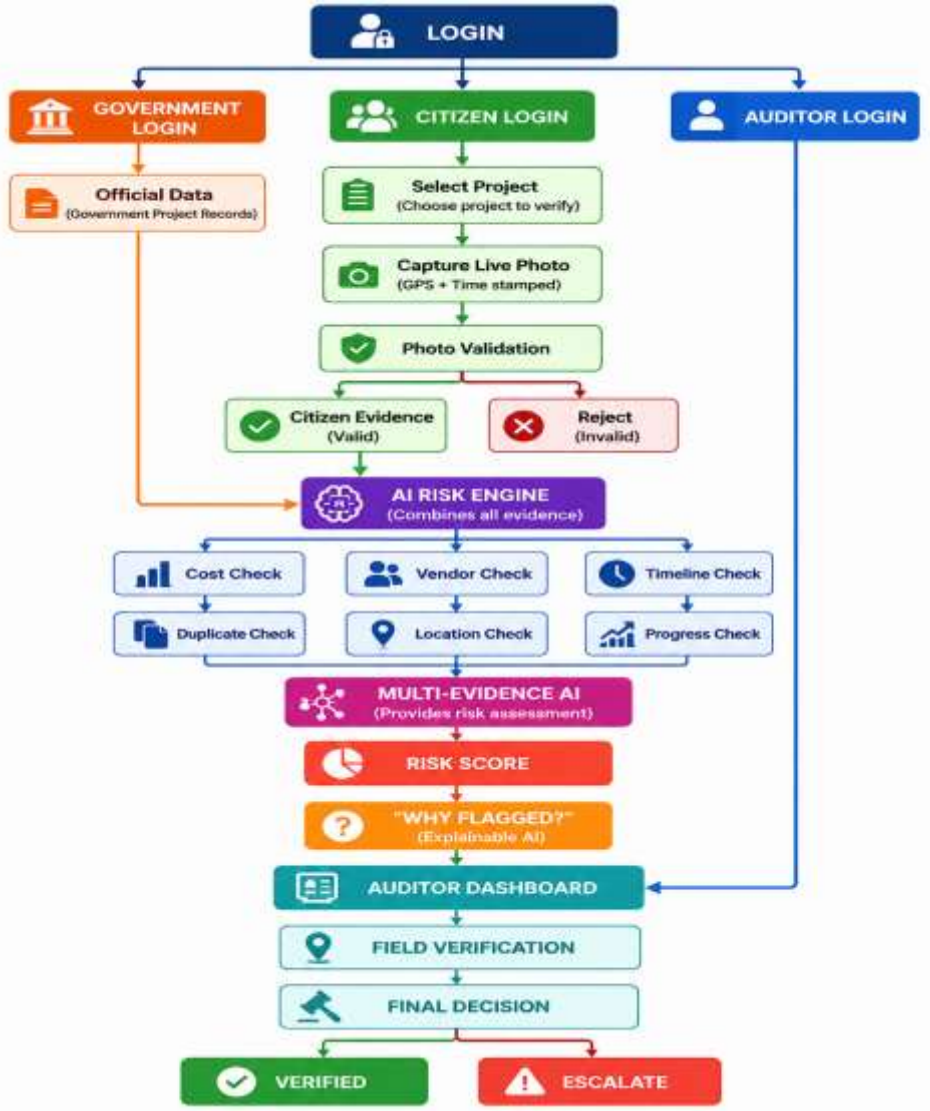
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IDEA TITLE

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Proposed Solution: MPLAD Project Guardian

- **AI-Powered Anomaly Detection** – Detects unusual cost, vendor, fund, timeline and progress patterns.
- **Citizen Ground Verification** – Collects live geotagged photos from project locations.
- **Multi-Evidence Verification** – Combines government records with citizen evidence.
- **Explainable AI Risk Scoring** – Generates risk scores and explains why a project was flagged.
- **Auditor Investigation Dashboard** – Helps auditors review flagged projects and supporting evidence.
- **Field Verification & Final Decision** – Enables physical verification and human-based final decisions.



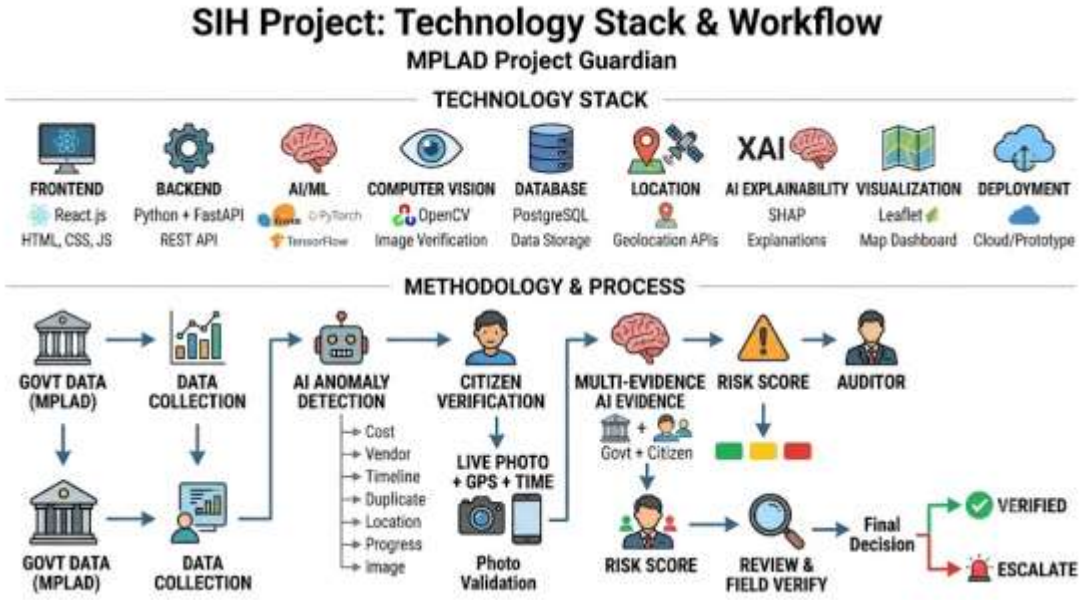
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TECHNICAL APPROACH

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Technologies to be Used

- **Frontend:** React.js / HTML, CSS, JavaScript
- **Backend:** Python + FastAPI
- **AI/ML:** Scikit-learn, PyTorch / TensorFlow
- **Computer Vision:** OpenCV for project image verification
- **Database:** PostgreSQL
- **Location Services:** GPS + Geolocation APIs
- **AI Explainability:** SHAP / feature-based explanations
- **Visualization:** Leaflet / Map-based dashboard
- **Deployment:** Cloud server / Local prototype



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FEASIBILITY AND VIABILITY

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❖ **Feasibility:**

1. Technical Feasibility:

--> AI-Based Anomaly Detection

--> Citizen-Based Geotag Verification

2. Operational Feasibility:

--> Existing Digital Infrastructure
--> Simple & Practical Workflow

3. Legal & Ethical Feasibility:

--> Data Privacy:
--> Human-in-the-Loop Decision:

❖ **Viability:**

1. Economic Viability:

→ Cost-Effective Monitoring

--> Affordable Technology

2. Social Viability:

--> Citizen Participation & Transparency
--> Faster & Better Auditing

3. Scalability:

--> District-to-Pan India Expansion
--> Future Multi-Scheme Application

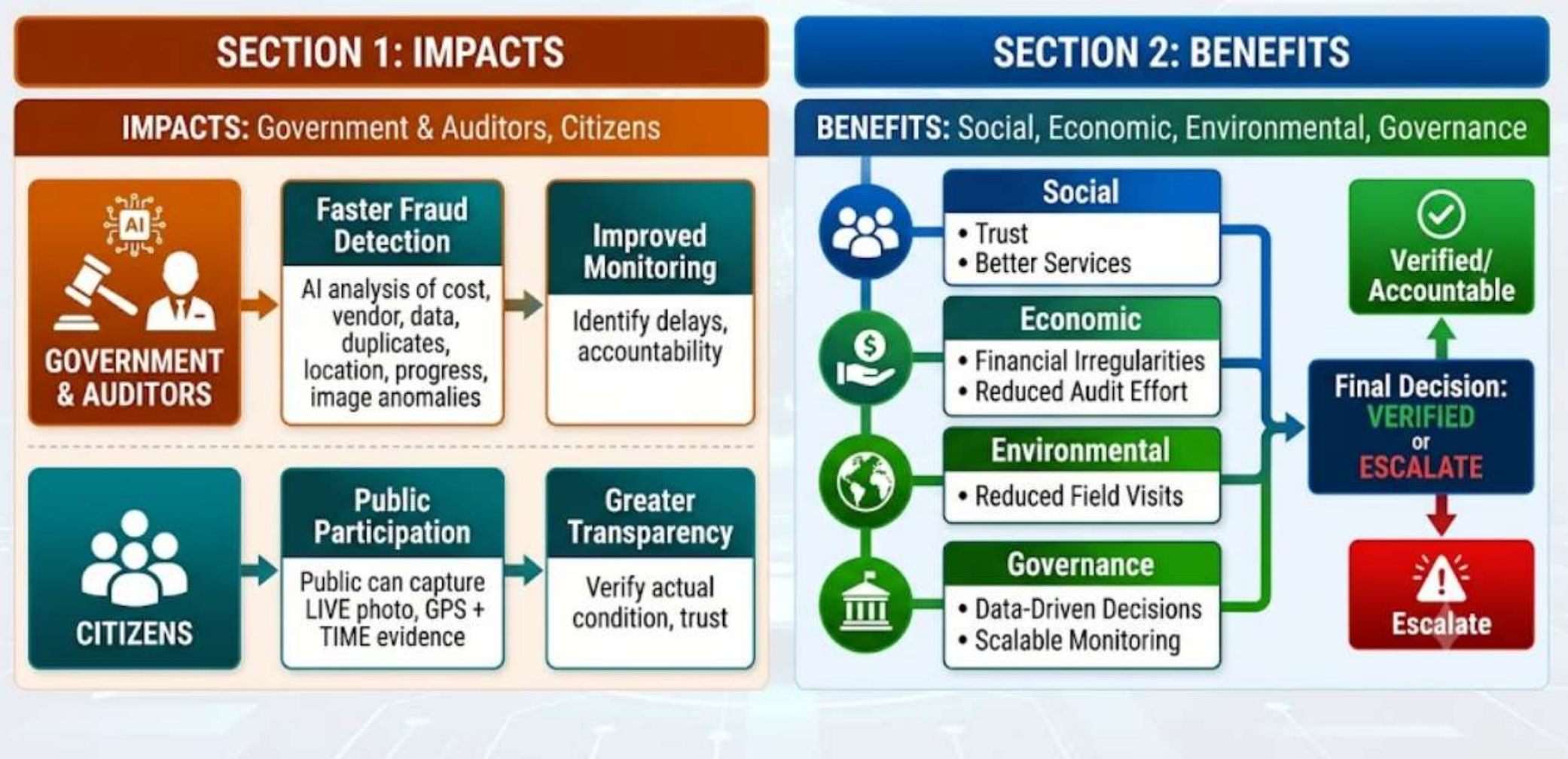
💡 Strategies to Overcome Challenges

Data Access → Use authorized APIs/data exports; use simulated datasets for the prototype.
Fake Images → Live camera capture + GPS + timestamp + image validation.
Poor Connectivity → Offline-first evidence capture with later synchronization.
AI Errors → Use explainable risk scoring and human auditor verification.
GPS Issues → Use a reasonable location-radius threshold and additional visual verification.
Privacy → Role-based access, encryption and minimum required data collection

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IMPACT AND BENEFITS

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RESEARCH AND REFERENCES

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1. MoSPI – MPLADS Official Portal
Understanding MPLADS implementation, monitoring and work-status workflow. (http://www.mplads.gov.in/)
2. e-SAKSHI Portal
Studied the existing end-to-end digital system for MPLADS implementation. Agencies. (https://en.vikaspedia.in/viewcontent/e-governance/online-citizen-services/mplads%E2%80%93esakshi-web-portal?lgn=en)
3. MPLADS Guidelines 2023
Understanding roles, project execution, monitoring and implementation procedures. (https://www.mospi.gov.in/about-us/mplads)
4. data.gov.in – MPLADS Dataset
Used fund, expenditure and work-status data as reference for anomaly detection. analysis. (https://www.data.gov.in/keywords/MPLADS)
5. MPLADS Public Dashboard
Studied project-level cost, status, timeline and location-related fields.
6. AI Research
Anomaly Detection + Explainable AI + Computer Vision + Geolocation Verification.